

Lectures on Monte Carlo Theory

Chapter 4: Simulation Output Analysis: Independent Replications

Based on Lorek & Rolski

Chapter Outline

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Introduction: Why Output Analysis? I

In Monte Carlo simulations, the result of running R independent replications is inherently random. Suppose we estimate $I = \pi$ using the sample average of $R = 1,000$ simulated values and obtain $\hat{I} = 3.14$. This is just one number—is it close to the truth? By how much could it be off? Without a rigorous answer, the simulation output is scientifically incomplete.

Output analysis is the discipline that makes simulation results scientifically meaningful. Its central tasks are:

- Construct a *confidence interval* that brackets the true value I with a specified probability $1 - \alpha$ (e.g. 95%).
- Determine how many replications R are needed to achieve a desired precision b (half-width of the confidence interval).

Introduction: Why Output Analysis? II

The Central Limit Theorem (CLT) provides the mathematical foundation: it guarantees that the sample mean is approximately normally distributed, so normal-distribution quantiles can be used to build confidence intervals.

The Central Goal of Chapter 4

Establish the **fundamental relationship**

$$b = \frac{z_{1-\alpha/2} D}{\sqrt{R}},$$

which links: the precision b (half-width of the CI), the number of replications R , and the variability constant D (which depends on the estimation problem). Any two of these quantities determine the third.

Introduction: Why Output Analysis? III

General Problem Setup

We wish to estimate a quantity I expressible as

$$I = \mathbb{E}Y,$$

where Y is a random variable with $\mathbb{E}Y^2 < \infty$ (finite variance $\sigma_Y^2 = \text{Var}(Y)$) that we can simulate. We draw R i.i.d. copies $Y_1, Y_2, \dots, Y_R$ and form the sample mean $\hat{Y}_R$.

Two fundamentally different notions of error arise throughout this chapter:

- The **absolute error** $|\hat{I} - I|$ measures the raw discrepancy between the estimate and the truth. We want it to be at most ε with high probability.
- The **relative error** $|\hat{I}/I - 1| = |\hat{I} - I|/I$ measures the discrepancy as a fraction of the true value. It is more informative when I itself is small (e.g. a rare-event probability).

Introduction: Why Output Analysis? IV

Error Control Conditions (for $\varepsilon, \delta \in (0, 1)$)

(E1) Absolute: $\mathbb{P}(|\hat{I} - I| \geq \varepsilon) \leq \delta$

(E2) Relative: $\mathbb{P}(|\hat{I}/I - 1| \geq \varepsilon) \leq \delta$

Equivalently: (E1') says $\mathbb{P}(|\hat{I} - I| \leq \varepsilon) \geq 1 - \delta$, and (E2') says $\mathbb{P}(|\hat{I} - I| \leq \varepsilon I) \geq 1 - \delta$. Note that (E2) is strictly stronger than (E1) when $I < 1$: it requires the error to be small *relative to* the true value.

This chapter focuses primarily on absolute error; relative error is revisited in Section 4.4, where we show that it is far harder to achieve when I is small.

4.1.1 The Sample Mean Estimator I

We want to compute $I = \mathbb{E}Y$ for a random variable Y we can simulate. The most natural estimator is the **sample mean** formed from R independent replications.

Sample Mean Estimator (Eq. 1.1)

$$\hat{Y}_R = \frac{1}{R} \sum_{j=1}^R Y_j,$$

where $Y_1, Y_2, \dots, Y_R$ are i.i.d. copies of Y . Each Y_j is one simulation run, and R is the total number of replications.

This estimator satisfies two fundamental properties:

Unbiasedness. $\mathbb{E}\hat{Y}_R = \mathbb{E}Y = I$ because expectation is linear. The expected value of the estimate equals the truth, so there is no systematic over- or underestimation.

4.1.1 The Sample Mean Estimator II

Strong consistency. By the Strong Law of Large Numbers, $\hat{Y}_R \rightarrow I$ with probability 1 as $R \rightarrow \infty$. This implies *weak consistency*: $\mathbb{P}(|\hat{Y}_R - I| \geq b) \rightarrow 0$ for every fixed $b > 0$. Weak consistency further implies that for every $b > 0$ and $0 < \alpha < 1$, there exists R_0 such that for all $R \geq R_0$:

$$\mathbb{P}(\hat{Y}_R - b < I \leq \hat{Y}_R + b) \geq 1 - \alpha. \quad (1.2)$$

This is the key consistency guarantee: for large enough R , the random interval $(\hat{Y}_R - b, \hat{Y}_R + b)$ contains the true value with high probability.

4.1.1 The Sample Mean Estimator III

Chebyshev-Based Error Bound (Remark 4.1.1)

Chebyshev's inequality states that for any random variable X with finite variance and for any $t > 0$: $\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \text{Var}(X)/t^2$. Applying this to $\hat{Y}_R$ (which has variance σ_Y^2/R) with $t = b_R$:

$$\mathbb{P}(|\hat{Y}_R - I| \geq b_R) \leq \frac{\sigma_Y^2/R}{b_R^2}.$$

Setting this bound equal to the significance level α and solving for b_R gives the *Chebyshev half-width*:

$$b_R^{\text{Ch}} = \frac{1}{\alpha^{1/2}} \cdot \frac{\sigma_Y}{\sqrt{R}}. \quad (1.3)$$

Here $\sigma_Y = \sqrt{\text{Var}(Y)}$ is the standard deviation of a single observation. Note that $b_R = O(R^{-1/2})$: the error decreases at the *Monte Carlo rate* regardless of the dimension of the problem.

The Chebyshev bound is valid for *any* distribution of Y , but it is quite conservative. The Central Limit Theorem gives a much sharper result at the cost of holding only approximately (for large R).

4.1.1 The Sample Mean Estimator IV

Central Limit Theorem (CLT, Eq. 1.4)

Let $Y_1, Y_2, \dots, Y_R$ be i.i.d. with $\sigma_Y = \sqrt{\text{Var } Y} < \infty$. As $R \rightarrow \infty$:

$$\mathbb{P}\left(\sum_{j=1}^R \frac{Y_j - I}{\sigma_Y \sqrt{R}} \leq x\right) \longrightarrow \Phi(x),$$

where $\Phi(x)$ is the standard normal CDF. Equivalently, $\sqrt{R}(\hat{Y}_R - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma_Y^2)$.

Let $z_{1-\alpha/2} = \Phi^{-1}(1 - \alpha/2)$ denote the $(1 - \alpha/2)$ -quantile of the standard normal. From the CLT:

$$\lim_{R \rightarrow \infty} \mathbb{P}\left(-z_{1-\alpha/2} < \frac{\hat{Y}_R - I}{\sigma_Y / \sqrt{R}} \leq z_{1-\alpha/2}\right) = 1 - \alpha.$$

Rearranging to isolate I gives the **fundamental relationship**:

4.1.1 The Sample Mean Estimator V

The Fundamental Relationship (Eq. 1.6)

$$b = z_{1-\alpha/2} \frac{\sigma_Y}{\sqrt{R}}, \quad (1.6)$$

where b is the CI half-width, $z_{1-\alpha/2}$ is the normal quantile, $\sigma_Y = \sqrt{\text{Var}(Y)}$ is the single-observation standard deviation, and R is the number of replications.

4.1.1 The Sample Mean Estimator VI

Required Number of Replications (Eq. 1.7)

Solving (1.6) for R : to achieve half-width b at confidence level $1 - \alpha$, we need

$$R = z_{1-\alpha/2}^2 \frac{\sigma_Y^2}{b^2}. \quad (1.7)$$

This is perhaps the most important formula in this chapter.

Common values. For $\alpha = 0.05$ (95% confidence): $z_{0.975} = 1.96$, giving $b = 1.96 \sigma_Y / \sqrt{R}$ and $R = 3.842 \sigma_Y^2 / b^2$. For $\alpha = 0.01$ (99% confidence): $z_{0.995} = 2.576$.

4.1.1 The Sample Mean Estimator VII

CLT vs. Chebyshev: Quantitative Comparison (Remark 4.1.5)

The ratio of the CLT half-width $b^{\text{CLT}} = z_{1-\alpha/2}\sigma_Y/\sqrt{R}$ to the Chebyshev half-width $b^{\text{Ch}} = \sigma_Y/(\alpha^{1/2}\sqrt{R})$ is:

$$\frac{b^{\text{CLT}}}{b^{\text{Ch}}} = z_{1-\alpha/2} \alpha^{1/2} = 1.96 \times \sqrt{0.05} = 0.4382 \quad (\alpha = 0.05).$$

Since the required sample size $R \propto b^{-2}$, the CLT-based formula needs only $(0.4382)^2 \approx 19.2\%$ as many replications as Chebyshev, at the same nominal confidence level. This is a huge practical saving. The price is that the CLT result is an approximation valid for large R , while Chebyshev is exact for any R .

4.1.1 The Confidence Interval in Practice I

In practice $\sigma_Y^2 = \text{Var}(Y)$ is unknown. We estimate it from the data using the **unbiased sample variance**.

Sample Variance (Two Definitions)

$$\hat{S}_R^2 = \frac{1}{R-1} \sum_{j=1}^R (Y_j - \hat{Y}_R)^2 \quad (\text{unbiased: } \mathbb{E}\hat{S}_R^2 = \sigma_Y^2)$$

$$\hat{\sigma}_R^2 = \frac{1}{R} \sum_{j=1}^R (Y_j - \hat{Y}_R)^2 \quad (\text{standard/biased})$$

The divisor $R-1$ in $\hat{S}_R^2$ makes it unbiased. For large R (typical in Monte Carlo), the two are nearly identical. Both are strongly consistent: $\hat{S}_R^2 \rightarrow \sigma_Y^2$ with probability 1.

4.1.1 The Confidence Interval in Practice II

Substituting $\hat{S}_R$ for σ_Y in (1.6) still works asymptotically by Slutsky's theorem. This gives the practical confidence interval:

$(1 - \alpha)$ -Confidence Interval (Eq. 1.9)

$$\left(\hat{Y}_R - z_{1-\alpha/2} \frac{\hat{S}_R}{\sqrt{R}}, \hat{Y}_R + z_{1-\alpha/2} \frac{\hat{S}_R}{\sqrt{R}} \right). \quad (1.9)$$

This interval contains the true value I with approximate probability $1 - \alpha$. The approximation improves as R grows. For 95% confidence with $\alpha = 0.05$: $z_{0.975} = 1.96$, so the CI becomes $\hat{Y}_R \pm 1.96 \hat{S}_R / \sqrt{R}$.

4.1.1 The Confidence Interval in Practice III

Recursive Update Formulas (Eqs. 1.11–1.12)

These allow online computation without storing all R observations. After observing Y_{R+1} , the updated mean and variance are:

$$\hat{Y}_{R+1} = \hat{Y}_R + \frac{Y_{R+1} - \hat{Y}_R}{R+1}, \quad (1.11)$$

$$\hat{\sigma}_{R+1}^2 = \hat{\sigma}_R^2 + (\hat{Y}_R)^2 - (\hat{Y}_{R+1})^2 + \frac{(Y_{R+1})^2 - \hat{\sigma}_R^2 - (\hat{Y}_R)^2}{R+1}. \quad (1.12)$$

Each new observation Y_{R+1} updates both statistics in $O(1)$ time and $O(1)$ memory. This is a streaming (online) algorithm, essential for very large simulations where storing all Y_j is impractical.

Practical workflow (pilot simulation):

- 1 Run a *pilot* simulation of size m (e.g. $m = 100$) to estimate σ_Y^2 via $\hat{S}_m^2$.
- 2 Compute the required main sample size $R = z_{1-\alpha/2}^2 \hat{S}_m^2 / b^2$ from (1.7).

4.1.1 The Confidence Interval in Practice IV

- ③ Run R replications; compute $\hat{Y}_R$ and $\hat{S}_R^2$.
- ④ Report the point estimate $\hat{Y}_R$ with the CI (1.9).

The pilot runs can be included in the main sample, or discarded; for large R it makes no difference.

4.1.1 Examples: Estimating π I

These examples illustrate how the *choice* of estimator Y (with $\mathbb{E}Y = \pi$) dramatically affects the required sample size, even though both estimators are unbiased for π .

Example 4.1.6 – Hit-or-Miss (Indicator) Estimator

Consider a unit square $[0, 1]^2$. A uniformly random point (U_1, U_2) falls inside the quarter-circle of radius 1 with probability $\pi/4$. Define $Y = 4 \cdot \mathbf{1}(U_1^2 + U_2^2 \leq 1)$ so that $\mathbb{E}Y = 4 \cdot (\pi/4) = \pi$. Since $\mathbf{1}(U_1^2 + U_2^2 \leq 1)$ is Bernoulli with parameter $\pi/4$:

$$\text{Var}(Y) = 16 \cdot \frac{\pi}{4} \left(1 - \frac{\pi}{4}\right) \approx 2.6968.$$

For first-digit accuracy ($b = 0.5$): $R \approx 1.96^2 \cdot 2.6968 / 0.25 \approx 42$. For second-digit accuracy ($b = 0.05$): $R \approx 4,144$ replications.

4.1.1 Examples: Estimating π II

Example 4.1.7 – Smooth (Madras) Estimator

The identity $\pi/4 = \int_0^1 \sqrt{1-x^2} dx$ suggests a smoother estimator. Define $Y = 4\sqrt{1-U^2}$ where $U \sim \mathcal{U}[0, 1]$:

$$\mathbb{E}Y = \int_0^1 4\sqrt{1-u^2} du = \pi.$$

Its variance is computed as $\mathbb{E}Y^2 - (\mathbb{E}Y)^2$:

$$\text{Var}(Y) = \int_0^1 16(1-u^2) du - \pi^2 = \frac{32}{3} - \pi^2 \approx 0.797.$$

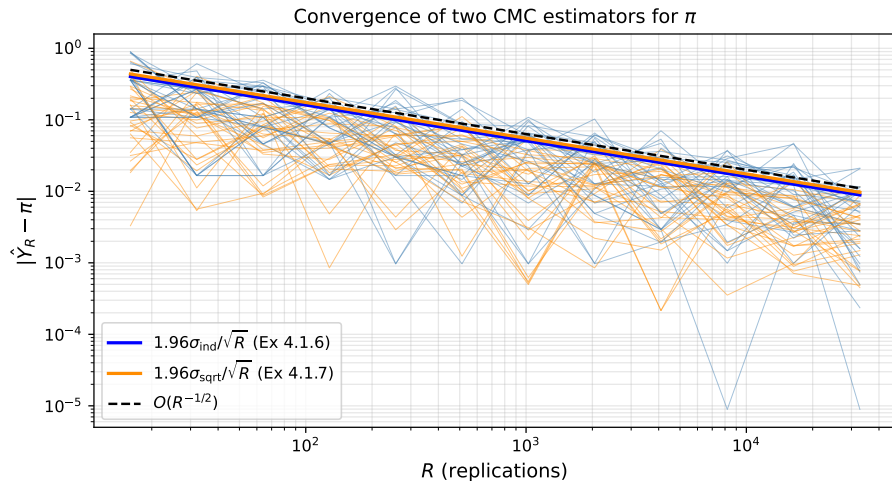
For first digit ($b = 0.5$): only $R \approx 12$ replications. For second digit ($b = 0.05$): $R \approx 1,224$ replications. This is approximately **3.38** $\times$ more efficient than the hit-or-miss estimator!

4.1.1 Examples: Estimating π III

Key Insight

Both estimators are unbiased for π , but the smooth one has $\sim 3.4\times$ lower variance. Since required $R \propto \text{Var}(Y)/b^2$, lower variance directly means fewer replications for the same precision. This motivates the variance reduction techniques of Chapter 5.

4.1.1 Examples: Estimating π IV



4.1.1 Crude Monte Carlo and Hit-or-Miss Integration I

Computing integrals is a central use case for Monte Carlo. Suppose we want:

$$I = \int_B k(x) dx,$$

where $B \subseteq \mathbb{R}^d$ is a bounded set with $\text{Vol}(B) = 1$ (rescale if needed; typically $B = [0, 1]^d$).

Crude Monte Carlo (CMC) Estimator (Eq. 1.13)

Let $V_1, \dots, V_R \sim \mathcal{U}(B)$ i.i.d. Setting $Y = k(V)$ ensures $\mathbb{E}Y = \int_B k(v) dv = I$. The estimator is:

$$\hat{Y}_R^{\text{CMC}} = \frac{1}{R} \sum_{j=1}^R k(V_j). \quad (1.13)$$

More generally, if X has density f on B and $Y = k(X)$, then $\mathbb{E}Y = \int_B k(x)f(x) dx = I$ and the CMC estimator is $\hat{Y}_R^{\text{CMC}} = \frac{1}{R} \sum_{j=1}^R k(X_j)$.

4.1.1 Crude Monte Carlo and Hit-or-Miss Integration II

Hit-or-Miss (HoM) Estimator (Example 4.1.11)

Assume $0 \leq k(x) \leq 1$ for all $x \in B = [0, 1]$. Generate $U_1 \sim \mathcal{U}[0, 1]$ and $U_2 \sim \mathcal{U}[0, 1]$ independently, and define $Y' = \mathbf{1}(k(U_1) > U_2)$. Then:

$$\mathbb{E}Y' = \int_0^1 \int_0^{k(u_1)} du_2 du_1 = \int_0^1 k(u_1) du_1 = I.$$

The HoM estimator is $\hat{Y}_R^{\text{HoM}} = \frac{1}{R} \sum Y'_j$. However, $\text{Var}(Y') = I(1 - I) \geq \text{Var}(k(U))$ in general, so HoM is less statistically efficient than CMC and requires twice as many uniform random numbers.

4.1.1 Crude Monte Carlo and Hit-or-Miss Integration III

Worked Example 4.1.10 (Madras)

Estimate $I = \int_0^1 k(x) dx$ with $k(x) = (e^x - 1)/(e - 1)$. Setting $Y = k(U)$, $U \sim \mathcal{U}[0, 1]$:

$$I = \mathbb{E}Y = \frac{1}{e-1} \left(\int_0^1 e^x dx - 1 \right) = \frac{e-2}{e-1} \approx 0.418.$$

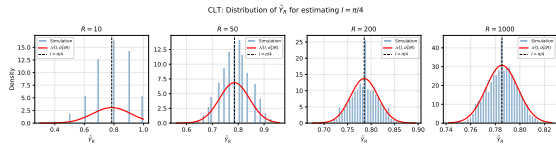
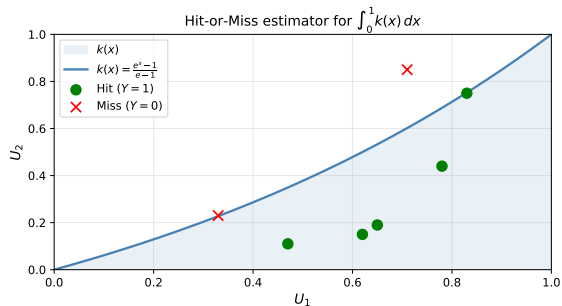
$$\mathbb{E}Y^2 = \int_0^1 k^2(x) dx = 0.2567.$$

$$\sigma_Y^2 = 0.2567 - (0.418)^2 = 0.0820.$$

To achieve precision b : $R = 0.3280/b^2$ replications needed.

For comparison, the HoM estimator $Y' = \mathbf{1}(k(U_1) > U_2)$ has $\text{Var}(Y') = I(1 - I) \approx 0.2433$. This is $0.2433/0.0820 \approx 2.97$ times the CMC variance, requiring about $3\times$ more replications for the same precision.

4.1.1 Crude Monte Carlo and Hit-or-Miss Integration IV



4.1.1 Bounding the Variance Without Knowing I

In practice, the true variance $\sigma_Y^2 = \text{Var}(Y)$ involves the unknown I . For $I = \int_B k(x)f(x) dx$ one can sometimes bound σ_Y^2 from above without knowing I , using only bounds on k .

Variance Bound for CMC (Remark 4.1.12)

For $Y = k(X)$ with $X \sim f$ on B :

$$\sigma_Y^2 = \int_B k^2(x)f(x) dx - \left(\int_B k(x)f(x) dx \right)^2.$$

If k can be bounded: $k_{\text{down}} \leq k(x) \leq k_{\text{up}}$, then $I \geq k_{\text{down}}$ and $\mathbb{E}Y^2 \leq k_{\text{up}}^2$, giving an upper bound on σ_Y^2 . In practice, only an upper bound on σ_Y^2 is needed; using it yields a slightly larger R than strictly necessary, but guarantees the desired confidence level.

4.1.1 Bounding the Variance Without Knowing I II

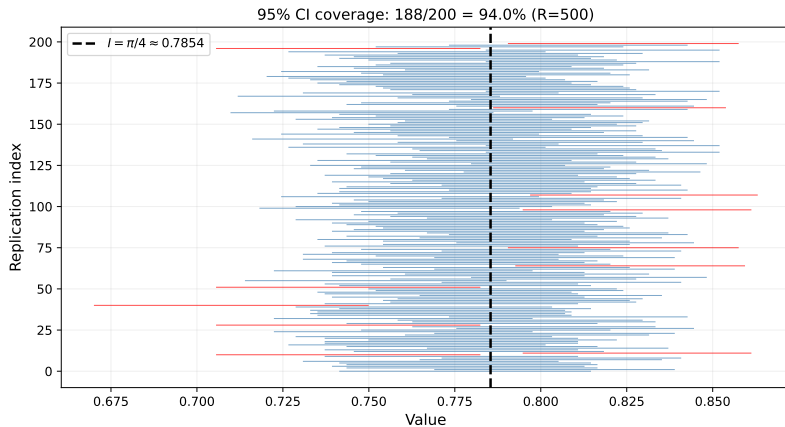
Bounding $\text{Var}(Y)$ for $k(x) = (e^x - 1)/(e - 1)$

Using $e \approx 2.71$ so that $e - 1 \approx 1.71$:

$$k_{\text{down}} = \int_0^1 \frac{2.71^x - 1}{1.72} dx \geq 0.4157,$$
$$\mathbb{E}Y^2 = \int_0^1 k^2(x) dx \leq \int_0^1 \frac{(2.72^2)^x - 2.71^x + 1}{(1.71)^2} dx =: k_{\text{up}}^2 = 0.2622.$$

Therefore $\sigma_Y^2 \leq 0.2622 - (0.4157)^2 = 0.0894$, close to the exact value 0.0820. Using this upper bound, one would compute $R \leq 0.0894 \times (1.96/b)^2$, which is only 9% more than the exact formula requires.

4.1.1 Bounding the Variance Without Knowing I III



4.1.1 Python: Computing a Confidence Interval

```
import numpy as np
from scipy import stats

def estimate_pi_ci(R, alpha=0.05, seed=42):
    """Estimate pi via hit-or-miss MC and report a (1-alpha) CI."""
    rng = np.random.default_rng(seed)
    U = rng.random((R, 2))
    #  $Y_j = 4 * 1(U_1^2 + U_2^2 \leq 1)$ 
    Y = 4.0 * (U[:, 0]**2 + U[:, 1]**2 <= 1).astype(float)

    Y_bar = Y.mean()           # sample mean -- estimator of pi
    S2     = Y.var(ddof=1)      # unbiased sample variance
    S      = np.sqrt(S2)

    z      = stats.norm.ppf(1 - alpha / 2)  #  $z_{\{1-\alpha/2\}}$ 
    margin = z * S / np.sqrt(R)             # half-width b

    return Y_bar, Y_bar - margin, Y_bar + margin

# Example: R=10000, alpha=0.05
Y_bar, lo, hi = estimate_pi_ci(R=10_000)
print(f"Estimate: {Y_bar:.4f}")
print(f"95% CI : ({lo:.4f}, {hi:.4f})")
print(f"True pi: {np.pi:.4f}")
# Typical output:
# Estimate: 3.1464
# 95% CI : (3.1142, 3.1786)
```

4.1.2 The Multidimensional Case I

Sometimes the quantity of interest is a vector $\mathbf{I} = (I^{(1)}, \dots, I^{(d)})^T = \mathbb{E}\mathbf{Y}$, where $\mathbf{Y} = (Y^{(1)}, \dots, Y^{(d)})^T \in \mathbb{R}^d$. For instance, we might want to estimate the mean height and weight of a population jointly, or estimate d different integrals simultaneously.

Vector Sample Mean Estimator

Given i.i.d. replications $\mathbf{Y}_1, \dots, \mathbf{Y}_R$:

$$\hat{\mathbf{Y}}_R = \frac{1}{R} \sum_{j=1}^R \mathbf{Y}_j = (\hat{Y}_R^{(1)}, \dots, \hat{Y}_R^{(d)})^T,$$

where each component $\hat{Y}_R^{(k)} = \frac{1}{R} \sum_{j=1}^R Y_j^{(k)}$ is the sample mean of the k -th coordinate. This is unbiased: $\mathbb{E}\hat{\mathbf{Y}}_R = \mathbf{I}$.

Let $\Sigma_{\mathbf{Y}} = \mathbb{E}[(\mathbf{Y} - \mathbf{I})(\mathbf{Y} - \mathbf{I})^T]$ be the $d \times d$ covariance matrix of $\mathbf{Y}$. The multivariate CLT states:

4.1.2 The Multidimensional Case II

Multivariate CLT (Eq. 1.23)

$$\sqrt{R}(\hat{\mathbf{Y}}_R - \mathbf{I}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathbf{Y}}).$$

This means $\hat{\mathbf{Y}}_R$ is approximately $\mathcal{N}(\mathbf{I}, \boldsymbol{\Sigma}_{\mathbf{Y}}/R)$ for large R .

4.1.2.1 Facts about the Multivariate Normal I

We recall two key facts about multivariate normal distributions needed to construct confidence regions. Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ a $d \times d$ nonsingular covariance matrix.

Spectral Decomposition of $\boldsymbol{\Sigma}$

Any symmetric positive-definite matrix $\boldsymbol{\Sigma}$ admits an eigendecomposition:

$$\boldsymbol{\Sigma} = \mathbf{F}\boldsymbol{\Lambda}\mathbf{F}^T = \sum_{i=1}^d \lambda_i \mathbf{f}_i \mathbf{f}_i^T,$$

where $\lambda_1, \dots, \lambda_d > 0$ are the eigenvalues and $\mathbf{f}_1, \dots, \mathbf{f}_d$ are orthonormal eigenvectors (columns of $\mathbf{F}$). The matrix square root is $\boldsymbol{\Sigma}^{1/2} = \mathbf{F}\boldsymbol{\Lambda}^{1/2}\mathbf{F}^T$. Two decompositions of $\boldsymbol{\Sigma}$ are important: (D1) Cholesky: $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$ (lower triangular $\mathbf{L}$); (D2) Square root: $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{1/2}\boldsymbol{\Sigma}^{1/2}$.

4.1.2.1 Facts about the Multivariate Normal II

Proposition 4.1.13 – Properties of Multivariate Normal

Let $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^T$. Then:

- Ⓐ $\mathbf{Z} = \mathbf{A}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{Id}_d)$. That is, the standardized vector has independent standard normal components.
- Ⓑ $(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi_d^2$. The squared Mahalanobis distance follows a chi-square distribution with d degrees of freedom.

Proof of (b). Since $\boldsymbol{\Sigma}^{-1} = (\mathbf{A}^T)^{-1}\mathbf{A}^{-1}$, we have $(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) = \mathbf{Z}^T \mathbf{Z} = \sum_{i=1}^d Z_i^2$ where $Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. This sum of squares of d independent standard normals is exactly χ_d^2 .

4.1.2.1 Facts about the Multivariate Normal III

Property (b) is the foundation for all multivariate confidence regions. Denoting by $q_{1-\alpha}(\chi_d^2)$ the $(1 - \alpha)$ -quantile of χ_d^2 :

$$\mathbb{P}\left((\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \leq q_{1-\alpha}(\chi_d^2)\right) = 1 - \alpha.$$

Quantile values at $\alpha = 0.05$ (so we need the 95th percentile):

d	$q_{0.95}(\chi_d^2)$	$\sqrt{q}$ (“effective radius”)	$d = 1$ recovers
1	3.841	1.960	$\pm 1.96\sigma$ interval
2	5.991	2.448	ellipse
5	11.07	3.327	5D ellipsoid
10	18.31	4.279	10D ellipsoid

For $d = 1$, $q_{0.95}(\chi_1^2) = z_{0.975}^2 = 1.96^2 = 3.841$, recovering the familiar one-dimensional 95% CI.

4.1.2.2 Shapes of Confidence Regions I

For $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$, a confidence region $\mathcal{C} \subset \mathbb{R}^d$ satisfying $\mathbb{P}(\mathbf{X} \in \mathcal{C}) \geq 1 - \alpha$ can take three shapes.

(1) Confidence Ellipsoid (Tightest, Eq. 1.15)

From Proposition 4.1.13(b), the set

$$\mathcal{C}^{\text{el}} = \{\mathbf{x} \in \mathbb{R}^d : \mathbf{x}^T \Sigma^{-1} \mathbf{x} \leq q_{1-\alpha}(\chi_d^2)\}$$

satisfies $\mathbb{P}(\mathbf{X} \in \mathcal{C}^{\text{el}}) = 1 - \alpha$ exactly. The boundary $\partial \mathcal{C}^{\text{el}}$ is a hyper-ellipsoid with principal axes $\sqrt{q_{1-\alpha}(\chi_d^2)} \lambda_i \mathbf{f}_i$, $i = 1, \dots, d$. The ellipsoid stretches most in the direction of largest variance. The ellipsoid is the *minimum-volume* region achieving coverage $1 - \alpha$.

4.1.2.2 Shapes of Confidence Regions II

(2) Confidence Parallelogram (Eq. 1.18)

Since $\Sigma^{-1/2}\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \mathbf{Id}_d)$, we can choose a hypercube $\mathcal{H} = (-z, z)^d$ with $z = \Phi^{-1}((\sqrt[d]{1-\alpha} + 1)/2)$, so that $\mathbb{P}(\Sigma^{-1/2}\mathbf{X} \in \mathcal{H}) = (2\Phi(z) - 1)^d = 1 - \alpha$. Transforming back gives the parallelogram:

$$\mathcal{C}^{\text{par}} = \Sigma^{1/2}\mathcal{H} + \boldsymbol{\mu}.$$

Its corners are $\Sigma^{1/2}\mathbf{v}_i + \boldsymbol{\mu}$ for $\mathbf{v}_i = (\pm z, \pm z, \dots, \pm z)$ (the 2^d corners of the cube).

4.1.2.2 Shapes of Confidence Regions III

(3) Rectangular Confidence Region (Conservative, Eq. 1.19)

A rectangular region $\mathcal{C}^{\text{rec}} = [-c_1, c_1] \times \cdots \times [-c_d, c_d]$ where $c_i = \sigma_i \Phi^{-1}((\sqrt[d]{1-\alpha} + 1)/2)$ and $\sigma_i = \sqrt{\Sigma(i,i)}$ is the marginal standard deviation, satisfies $\mathbb{P}(\mathbf{X} \in \mathcal{C}^{\text{rec}}) \geq 1 - \alpha$ (by the Šidák inequality). This is conservative because it ignores correlations, but easy to compute and interpret as axis-aligned intervals.

Example 4.1.14: $\Sigma = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$, $\alpha = 0.05$

Eigenvalues: $\lambda_1 = (7 + \sqrt{41})/2 \approx 6.7015$, $\lambda_2 = (7 - \sqrt{41})/2 \approx 0.2984$. Eigenvectors:

$\mathbf{f}_1 = (-0.3310, -0.9436)^T$, $\mathbf{f}_2 = (-0.9436, 0.3310)^T$.

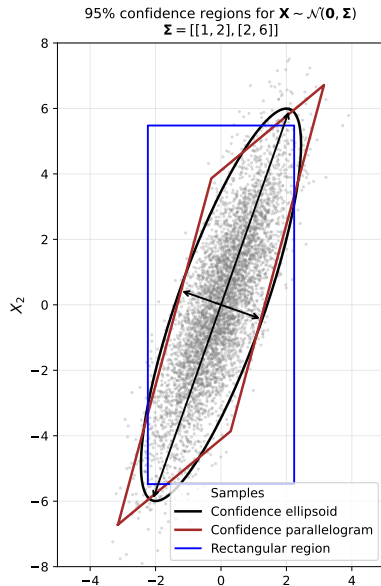
$q_{0.95}(\chi_2^2) = 5.991$. Ellipsoid axis half-lengths: $\sqrt{5.991 \times 6.7015} \approx 6.337$ and $\sqrt{5.991 \times 0.2984} \approx 1.337$.

Parallelogram: $z = \Phi^{-1}((\sqrt{0.95} + 1)/2) = 2.2365$; corners $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 = (\pm z, \pm z)^T$ mapped by $\Sigma^{1/2}$.

Rectangle: $c_1 = 1 \times 2.2365 = 2.2365$, $c_2 = \sqrt{6} \times 2.2365 \approx 5.478$. Exact rectangle with $z = 2.146$:

$c_1^{\text{exact}} = 2.146$, $c_2^{\text{exact}} = 5.258$.

4.1.2.2 Shapes of Confidence Regions IV



4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$

In practice we do not observe $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{Y}})$ directly. Instead, the multivariate CLT gives $\hat{\mathbf{Y}}_R \approx \mathcal{N}(\mathbf{I}, \Sigma_{\mathbf{Y}}/R)$. Replacing Σ by $\Sigma_{\mathbf{Y}}/R$ in the confidence regions of the previous section, the ellipsoid becomes centered at $\hat{\mathbf{Y}}_R$ with principal axes of half-length $\sqrt{q_{1-\alpha}(\chi_d^2) \lambda_i / R}$ in direction $\mathbf{f}_i$ (Eq. 1.24).

Since $\Sigma_{\mathbf{Y}}$ is unknown, we substitute the **sample covariance matrix**:

$$\hat{\mathbf{S}}_R^2 = \frac{1}{R-1} \sum_{k=1}^R (\mathbf{Y}_k - \hat{\mathbf{Y}}_R)(\mathbf{Y}_k - \hat{\mathbf{Y}}_R)^T. \quad (1.25)$$

This is the unbiased estimator of $\Sigma_{\mathbf{Y}}$.

4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$ II

Procedure: Estimated Confidence Ellipsoid

- 1 Compute sample mean $\hat{\mathbf{Y}}_R$ and sample covariance $\hat{\mathbf{S}}_R^2$.
- 2 Compute eigenvalues λ_i and eigenvectors $\mathbf{f}_i$ of $\hat{\mathbf{S}}_R^2$.
- 3 The $1 - \alpha$ confidence ellipsoid is centered at $\hat{\mathbf{Y}}_R$ with principal axes of half-length $\sqrt{q_{1-\alpha}(\chi_d^2)} \lambda_i / R$ in direction $\mathbf{f}_i$.

Hotelling's T^2 (Remark 4.1.15). When $\Sigma_{\mathbf{Y}}$ is replaced by $\hat{\mathbf{S}}_R^2$, the statistic $R(\hat{\mathbf{Y}}_R - \mathbf{I})^T (\hat{\mathbf{S}}_R^2)^{-1} (\hat{\mathbf{Y}}_R - \mathbf{I})$ follows Hotelling's T^2 -distribution. For large R this converges to χ_d^2 , justifying the chi-square quantiles in practice.

4.1.2.3 Confidence Regions for the Estimator $\hat{\mathbf{Y}}_R$ III

Example 4.1.16 – Heights and Weights of 18-year-olds ($R = 1000$, $d = 2$)

Using the SOCR HeightsWeights dataset (first 1000 entries). Component 1: height (inches), Component 2: weight (pounds).

Individual 95% confidence intervals:

$$\hat{Y}_R^{(1)} = 127.48, \hat{S}_R^{(1)} = 11.80 : \quad \text{CI for height: } (126.75, 128.21)$$

$$\hat{Y}_R^{(2)} = 68.05, \hat{S}_R^{(2)} = 1.94 : \quad \text{CI for weight: } (67.93, 68.17)$$

Sample covariance and eigendecomposition:

$$\hat{\mathbf{S}}_R^2 = \begin{pmatrix} 139.29 & 11.40 \\ 11.40 & 3.75 \end{pmatrix}, \quad \lambda_1 = 140.24, \lambda_2 = 2.79.$$

$\mathbf{f}_1 \approx (0.9965, 0.0832)^T$ (mostly weight direction), $\mathbf{f}_2 \approx (-0.0832, 0.9965)^T$. At $\alpha = 0.05$: $q_{0.95}(\chi_2^2) = 5.991$. Ellipsoid axis half-lengths: $\sqrt{5.991 \times 140.24/1000} \approx 0.916$ and $\sqrt{5.991 \times 2.79/1000} \approx 0.129$. The confidence ellipse is extremely elongated along the weight axis.

4.1.2 Python: Multivariate Confidence Ellipsoid

```
import numpy as np
from scipy.stats import chi2

def conf_ellipsoid(Y_samples, alpha=0.05):
    """Confidence ellipsoid for the mean of d-dimensional i.i.d. samples."""
    R, d = Y_samples.shape
    Y_bar = Y_samples.mean(axis=0)           # sample mean (d-vector)

    diff = Y_samples - Y_bar
    S2 = (diff.T @ diff) / (R - 1)          # sample covariance matrix
    eigvals, eigvecs = np.linalg.eigh(S2)   # eigendecomposition
    q = chi2.ppf(1 - alpha, df=d)          # chi-square quantile

    # Half-lengths of principal axes of the confidence ellipsoid
    axis_lengths = np.sqrt(q * eigvals / R)
    return Y_bar, axis_lengths, eigvecs, S2

# Simulate 2D example: I = (2.0, 3.0), Sigma = [[1, 0.5], [0.5, 2]]
rng = np.random.default_rng(42)
Sigma = np.array([[1., 0.5], [0.5, 2.]])
L = np.linalg.cholesky(Sigma)
R = 1000
Y = (L @ rng.standard_normal((2, R))).T + np.array([2.0, 3.0])

Y_bar, axes, vecs, S2 = conf_ellipsoid(Y, alpha=0.05)
print(f"Estimated mean      : {Y_bar}")
print(f"True mean             : [2.0, 3.0]")
```

4.2 Estimation of $k(I)$: Motivation I

In many applications the quantity of interest is not $I = \mathbb{E}Y$ itself, but a smooth transformation $z = k(I)$ of the mean. For example:

- **Insurance net premium:** $\Pi = r \mathbb{E}e^{-rT} / (1 - \mathbb{E}e^{-rT}) = k(I)$ where $I = \mathbb{E}e^{-rT}$ and T is the insured's lifetime.
- **Standard deviation:** $k(I) = \sqrt{I}$ where $I = \mathbb{E}Y^2 - (\mathbb{E}Y)^2$.
- **Ratio of expectations:** $z = I^{(1)} / I^{(2)}$, e.g. expected cost per unit of expected throughput.

4.2 Estimation of $k(I)$: Motivation II

Problem Setup and Natural Estimator

Let $k : \mathbb{R}^d \rightarrow \mathbb{R}$ be a prescribed continuous function, $\mathbf{I} = \mathbb{E}\mathbf{Y}$, and let $\hat{\mathbf{X}}_R$ be a strongly consistent estimator of $\mathbf{I}$. The natural estimator of $z = k(\mathbf{I})$ is:

$$\hat{Z}_R = k(\hat{\mathbf{X}}_R).$$

The **asymptotic variance** (or Time Average Variance Constant, TAVC) is:

$$\zeta^2 = \lim_{R \rightarrow \infty} R \operatorname{Var}(\hat{Z}_R).$$

Important caveat: $k(\hat{X}_R)$ is generally biased. Even when $\hat{X}_R$ is unbiased for I , the estimator $k(\hat{X}_R)$ is in general biased for $k(I)$ because of the nonlinearity of k . For example, $\mathbb{E}[\hat{Y}_R^2] = (\mathbb{E}Y)^2 + R^{-1}\sigma_Y^2 \neq (\mathbb{E}Y)^2$ when $k(y) = y^2$. However, the bias is $O(R^{-1})$ while the standard deviation is $O(R^{-1/2})$, so for large R the bias is negligible relative to the uncertainty.

4.2.1 The Delta Method (Case $\hat{X}_R = \hat{Y}_R$) I

The key tool for handling the transformation k is the **delta method**. The idea is simple: if k is smooth and $\hat{Y}_R \approx I$ for large R , then we can linearize $k(\hat{Y}_R)$ around I using a Taylor expansion.

Taylor Expansion Argument

If k is continuously differentiable near I :

$$k(x) - k(I) = k'(I)(x - I) + o(x - I) \quad \text{as } x \rightarrow I.$$

Substituting $x = \hat{Y}_R$:

$$\sqrt{R}(k(\hat{Y}_R) - k(I)) = k'(I) \cdot \underbrace{\sqrt{R}(\hat{Y}_R - I)}_{\rightarrow \mathcal{N}(0, \sigma_Y^2)} + \underbrace{\sqrt{R} \cdot o(\hat{Y}_R - I)}_{\rightarrow 0 \text{ in prob.}}.$$

The error term vanishes in probability because $\sqrt{R}|\hat{Y}_R - I| = O_p(1)$ (bounded in probability by the CLT) while $o(|\hat{Y}_R - I|)/|\hat{Y}_R - I| \rightarrow 0$.

4.2.1 The Delta Method (Case $\hat{X}_R = \hat{Y}_R$) II

Proposition 4.2.1 – Delta Method (1D)

Let $Y_1, Y_2, \dots$ be i.i.d., $\hat{Y}_R = \frac{1}{R} \sum Y_j$, $I = \mathbb{E}Y$, $\sigma_Y^2 = \text{Var}(Y)$, and k continuously differentiable at I . Then:

$$\sqrt{R}(k(\hat{Y}_R) - k(I)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2), \quad \varsigma^2 = k'(I)^2 \sigma_Y^2. \quad (2.2)$$

Here $k'(I)$ is the derivative of k at the true value I , $\sigma_Y^2 = \text{Var}(Y)$, and ς^2 is the asymptotic variance of the transformed estimator.

4.2.1 The Delta Method (Case $\hat{X}_R = \hat{Y}_R$) III

Since I and σ_Y^2 are unknown, we replace them by $\hat{Y}_R$ and $\hat{S}_R^2$ respectively (both are consistent), giving:

Practical CI for $k(I)$ at Level $1 - \alpha$ (Eq. 2.3)

$$\left(k(\hat{Y}_R) - z_{1-\alpha/2} \frac{\hat{S}_R |k'(\hat{Y}_R)|}{\sqrt{R}}, k(\hat{Y}_R) + z_{1-\alpha/2} \frac{\hat{S}_R |k'(\hat{Y}_R)|}{\sqrt{R}} \right). \quad (2.3)$$

The estimated asymptotic standard deviation is $\hat{S}_R |k'(\hat{Y}_R)|/\sqrt{R}$. This CI is asymptotically valid: its coverage converges to $1 - \alpha$.

4.2.1 The Delta Method (Case $\hat{X}_R = \hat{Y}_R$) IV

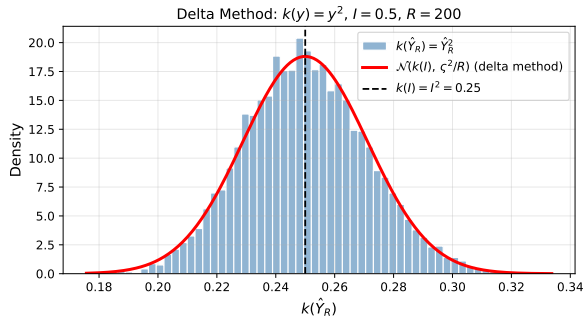
Example 4.2.2 – Life Insurance Net Premium

The net premium for whole-life assurance at interest rate r is $\Pi = r \mathbb{E}e^{-rT} / (1 - \mathbb{E}e^{-rT}) = k(I)$, where $I = \mathbb{E}e^{-rT}$ and $T = T_x$ is the future lifetime. We set $Y = e^{-rT}$ so $\hat{Y}_R = \frac{1}{R} \sum e^{-rT_j}$ estimates I . The function is $k(x) = rx/(1-x)$ with derivative $k'(x) = r/(1-x)^2$. The confidence interval (2.3) becomes:

$$k(\hat{Y}_R) \pm \frac{z_{1-\alpha/2} r \hat{S}_R}{(1 - \hat{Y}_R)^2 \sqrt{R}}.$$

Notice that the derivative factor $r/(1 - \hat{Y}_R)^2$ amplifies the uncertainty: as $\hat{Y}_R \rightarrow 1$ (long expected lifetime or high interest rate), the CI widens.

4.2.1 The Delta Method (Case $\hat{X}_R = \hat{Y}_R$) V



4.2.1 Multidimensional Delta Method I

When $\mathbf{I} = \mathbb{E}\mathbf{Y}$ is d -dimensional and $k : \mathbb{R}^d \rightarrow \mathbb{R}$, the delta method generalizes by using the gradient of k .

Two-Dimensional Case (Proposition 4.2.3, Eq. 2.6)

Let $\mathbf{Y}_j = (Y_j^{(1)}, Y_j^{(2)})^T$ be i.i.d., $\mathbf{I} = (I^{(1)}, I^{(2)})^T$, and $k(y_1, y_2)$ continuously differentiable at $\mathbf{I}$. Then:

$$\sqrt{R}(k(\hat{\mathbf{Y}}_R) - k(\mathbf{I})) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2),$$

where the asymptotic variance involves the gradient evaluated at $\mathbf{I}$:

$$\varsigma^2 = (k'_{y_1})^2 \Sigma_{\mathbf{Y}}(1, 1) + (k'_{y_2})^2 \Sigma_{\mathbf{Y}}(2, 2) + 2k'_{y_1} k'_{y_2} \Sigma_{\mathbf{Y}}(1, 2), \quad (2.6)$$

with $k'_{y_i} = \partial k / \partial y_i|_{\mathbf{y}=\mathbf{I}}$.

4.2.1 Multidimensional Delta Method II

General d -Dimensional Case (Proposition 4.2.8, Eq. 2.15)

If $k : \mathbb{R}^d \rightarrow \mathbb{R}$ is continuously differentiable at $\mathbf{I}$ and $\sqrt{R}(\hat{\mathbf{Y}}_R - \mathbf{I}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \Sigma)$, then:

$$\varsigma^2 = \nabla k(\mathbf{I}) \Sigma_{\mathbf{Y}} (\nabla k(\mathbf{I}))^T = \sum_{i,j=1}^d k^{(i)}(\mathbf{I}) k^{(j)}(\mathbf{I}) \Sigma_{\mathbf{Y}}(i, j), \quad (2.15)$$

where $k^{(i)}(\mathbf{I}) = \partial k / \partial x_i|_{\mathbf{x}=\mathbf{I}}$. This is a quadratic form in the gradient of k at $\mathbf{I}$.

4.2.1 Multidimensional Delta Method III

In practice $\mathbf{I}$ and $\Sigma_{\mathbf{Y}}$ are estimated by $\hat{\mathbf{Y}}_R$ and $\hat{\mathbf{S}}_R^2$. The estimated asymptotic variance is (Eq. 2.16):

$$\hat{\varsigma}^2 = \sum_{i,j=1}^d k^{(i)}(\hat{\mathbf{Y}}_R) k^{(j)}(\hat{\mathbf{Y}}_R) \hat{\mathbf{S}}_R^2(i, j).$$

The practical CI at level $1 - \alpha$ for $z = k(\mathbf{I})$ is:

$$k(\hat{\mathbf{Y}}_R) \pm z_{1-\alpha/2} \frac{\hat{\varsigma}}{\sqrt{R}} \quad (\text{Eq. 2.8})$$

4.2.1 Multidimensional Delta Method IV

Example 4.2.5 – Ratio of Expectations

We want $z = I^{(1)}/I^{(2)}$, estimated by $\hat{Z}_R = \hat{Y}_R^{(1)}/\hat{Y}_R^{(2)}$. With $k(y_1, y_2) = y_1/y_2$: partial derivatives $k'_{y_1} = 1/y_2$ and $k'_{y_2} = -y_1/y_2^2$. The asymptotic variance simplifies to:

$$\varsigma^2 = \frac{\mathbb{E}(Y^{(1)} - zY^{(2)})^2}{(I^{(2)})^2},$$

which can be estimated from the data as (Eq. 2.13):

$$\widehat{\text{Var}}(\hat{Z}_R) = \frac{1}{R^2(\hat{Y}_R^{(2)})^2} \sum_{i=1}^R (Y_i^{(1)} - \hat{Z}_R Y_i^{(2)})^2.$$

4.2.2 General Consistent Estimator $\hat{X}_R$ I

The delta method extends beyond the sample mean. Consider any strongly consistent estimator $\hat{X}_R \rightarrow I$ satisfying two asymptotic assumptions:

- ① $\sqrt{R}(\hat{X}_R - \mathbb{E}\hat{X}_R) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$ with $0 < \sigma^2 < \infty$ (asymptotic normality of centered estimator).
- ② $\sqrt{R}(\mathbb{E}\hat{X}_R - I) \rightarrow 0$ as $R \rightarrow \infty$ (the bias vanishes faster than $1/\sqrt{R}$, i.e. the estimator is asymptotically unbiased).

Under (A1) and (A2): $\sqrt{R}(\hat{X}_R - I) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$. For $\hat{Z}_R = k(\hat{X}_R)$, the delta method gives:

$$\sqrt{R}(k(\hat{X}_R) - k(I)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2), \quad \varsigma^2 = k'(I)^2 \sigma^2.$$

4.2.2 General Consistent Estimator $\hat{X}_R$ II

Asymptotic Bias of $\hat{Z}_R$ (Eq. 2.17)

Under the stronger assumption (A2'): $R(\mathbb{E}\hat{X}_R - I) \rightarrow \beta^X$ for some finite β^X , the R^{-1} -order asymptotic bias of $\hat{Z}_R = k(\hat{X}_R)$ is:

$$\beta^Z = \beta^X k'(I) + \frac{\sigma^2 k''(I)}{2}. \quad (2.17)$$

The first term $\beta^X k'(I)$ propagates the bias of $\hat{X}_R$ through k . The second term $\sigma^2 k''(I)/2$ is the *Jensen's inequality effect*: the curvature of k causes bias even when $\hat{X}_R$ is unbiased ($\beta^X = 0$). A bias-corrected estimator is $k(\hat{X}_R) - \beta^Z/R$, but this correction is usually negligible because the std dev $\varsigma/\sqrt{R}$ dominates the bias β^Z/R for large R .

4.2.3 Mean Square Error, Bias, and Variance I

The **mean square error** (MSE) is a single number that captures both the random fluctuations and systematic bias of an estimator.

Bias-Variance Decomposition (Eq. 2.19)

For estimator $\hat{X}_R$ of I :

$$\text{MSE}(\hat{X}_R) = \mathbb{E}(\hat{X}_R - I)^2 = \underbrace{\text{Var}(\hat{X}_R)}_{\text{variance}} + \underbrace{(\mathbb{E}\hat{X}_R - I)^2}_{\text{squared bias}}.$$

The *variance* term captures random fluctuation around the estimator's own mean; the *squared bias* term captures systematic over- or underestimation. An unbiased estimator has $\text{MSE} = \text{Var}$.

For large R , using $\lim_{R \rightarrow \infty} R \text{Var}(\hat{X}_R) = \varsigma^2$ and $\lim_{R \rightarrow \infty} R(\mathbb{E}\hat{X}_R - I) = \beta$:

$$\text{MSE}(\hat{X}_R) \approx \frac{\varsigma^2}{R} + \frac{\beta^2}{R^2}.$$

4.2.3 Mean Square Error, Bias, and Variance II

The variance term ς^2/R is of order R^{-1} (the Monte Carlo rate), while the squared bias β^2/R^2 is of order R^{-2} , which is *negligible* for large R . This justifies ignoring bias corrections in typical Monte Carlo settings (Remark 4.2.4).

4.2.3 Mean Square Error, Bias, and Variance III

Example 4.2.9 – $k(y) = y^2$

Let $\hat{Z}_R = \hat{Y}_R^2$ where $Y_1, \dots, Y_R$ are i.i.d. with $I = \mathbb{E}Y$. Then $k(y) = y^2$, $k'(y) = 2y$, $k''(y) = 2$. Since $\hat{Y}_R$ is unbiased ($\beta^X = 0$), formula (2.17) gives the asymptotic bias of order R^{-1} :

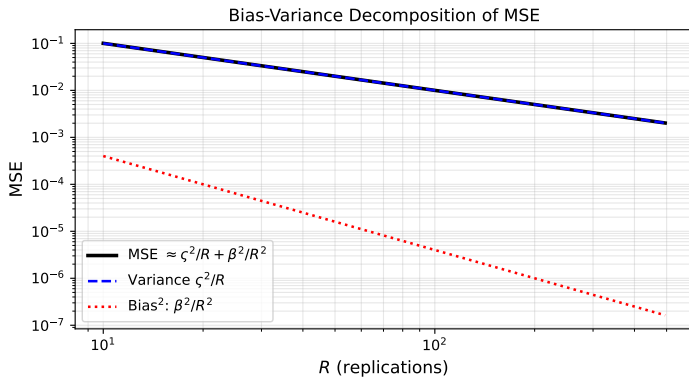
$$\beta^Z = 0 + \frac{\sigma_Y^2 \cdot 2}{2} = \sigma_Y^2.$$

We can verify directly from the identity $\mathbb{E}(S^2) = (\mathbb{E}S)^2 + \text{Var}(S)$ applied to $S = \hat{Y}_R$:

$$\mathbb{E}(\hat{Y}_R)^2 = (\mathbb{E}Y)^2 + \text{Var}(\hat{Y}_R) = k(I) + \frac{\sigma_Y^2}{R}.$$

So the bias is σ_Y^2/R , confirming formula (2.17). The delta method asymptotic variance is $\varsigma^2 = (2I)^2 \sigma_Y^2 = 4I^2 \sigma_Y^2$. The bias-corrected estimator is $\hat{Y}_R^2 - \hat{S}_R^2/R$, but for large R the correction is tiny compared to the CI width.

4.2.3 Mean Square Error, Bias, and Variance IV



4.2 Python: Delta Method Confidence Interval

```
import numpy as np
from scipy import stats

def delta_method_ci(Y_samples, k_func, k_prime, alpha=0.05):
    """
    Compute CI for  $k(I)$  using the delta method.
    Y_samples : 1D array of i.i.d. realizations of  $Y$  ( $E[Y]=I$ )
    k_func     : callable, the transformation  $k$ 
    k_prime    : callable, derivative  $k'$ 
    """
    R      = len(Y_samples)
    Y_bar  = Y_samples.mean()
    S      = Y_samples.std(ddof=1)      # sample std dev

    z_est  = k_func(Y_bar)              # point estimate
    asymp_std = abs(k_prime(Y_bar)) * S / np.sqrt(R) # asymptotic std dev

    z_q = stats.norm.ppf(1 - alpha / 2)
    return z_est, z_est - z_q * asymp_std, z_est + z_q * asymp_std

# Example:  $k(y) = y^2$ .  $Y \sim N(0.5, 0.09)$ ,  $I=0.5$ ,  $k(I)=0.25$ .
rng = np.random.default_rng(42)
Y = rng.normal(0.5, 0.3, size=1000)
est, lo, hi = delta_method_ci(
    Y, k_func=lambda y: y**2, k_prime=lambda y: 2*y)
print(f"k(I) estimate: {est:.4f}")
print(f"95% CI: ({lo:.4f}, {hi:.4f})")
```

4.3 Estimators of a Quantile: Setup I

So far we have estimated *means* $I = \mathbb{E}X$. Quantiles are a different summary statistic. The p -th quantile of a distribution is the value below which a fraction p of the probability lies. In financial risk management, the quantile is called **Value-at-Risk (VaR)**.

Definition: p -th Quantile

Let $X \sim F$ (with continuous distribution function F and density f). The p -th quantile for $p \in (0, 1)$ is the generalized inverse:

$$q_p = F^{\leftarrow}(p) = \inf\{x : F(x) \geq p\}.$$

For a continuous F , this is the unique solution to $F(q_p) = p$. In financial mathematics: $q_p = \text{VaR}_X(p)$.

Given an i.i.d. sample $X_1, \dots, X_R$ from F , the natural estimator is the **sample quantile** (order statistic). Denoting the ordered observations $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(R)}$, the empirical quantile function is (Eq. 3.1):

$$\hat{F}_R^{\leftarrow}(p) = \begin{cases} X_{(\lfloor pR \rfloor)} & \text{if } pR \in \mathbb{Z}, \\ X_{(\lfloor pR \rfloor + 1)} & \text{otherwise.} \end{cases}$$

4.3 Estimators of a Quantile: Setup II

For asymptotic analysis we work with the simplified estimator $\hat{q}_p = X_{(\lfloor pR \rfloor)}$. The difference $|\hat{F}_R^{\leftarrow}(p) - \hat{q}_p|$ converges to 0 as $R \rightarrow \infty$, so both are asymptotically equivalent.

4.3 Asymptotic Normality of the Sample Quantile I

Proposition 4.3.1 – Asymptotic Normality (Eq. 3.2)

Assume f is continuous and positive in a neighbourhood of q_p . Then:

(i) $\hat{q}_p = X_{(\lfloor pR \rfloor)}$ is a strongly consistent estimator of q_p .

(ii) $\sqrt{R}(\hat{q}_p - q_p) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \varsigma^2)$, where $\varsigma^2 = \frac{p(1-p)}{f(q_p)^2}$.

The asymptotic variance ς^2 is large when $f(q_p)$ is small (the distribution is flat near the quantile, making it hard to locate precisely) and small when $f(q_p)$ is large (sharp peak near q_p).

Proof sketch (two key steps):

Step 1. The $\lfloor Rp \rfloor$ -th order statistic of R i.i.d. $\mathcal{U}[0, 1]$ variables has a $\text{Beta}(\lfloor Rp \rfloor, R - \lfloor Rp \rfloor)$ distribution (Lemma 4.3.2). Using the representation as ratios of gamma random variables and the CLT for Erlang variables, one shows:

$$\sqrt{R}(U_{\lfloor Rp \rfloor} - p) \xrightarrow{\mathcal{D}} \mathcal{N}(0, p(1-p)).$$

4.3 Asymptotic Normality of the Sample Quantile II

Step 2. Since $X_{(\lfloor Rp \rfloor)} = F^{\leftarrow}(U_{\lfloor Rp \rfloor})$, apply the quantile function's Taylor expansion:
 $F^{\leftarrow}(p+x) \approx F^{\leftarrow}(p) + x/f(q_p)$. This rescales the variance by $(1/f(q_p))^2$, giving $\varsigma^2 = p(1-p)/f(q_p)^2$.

4.3 Asymptotic Normality of the Sample Quantile III

Worked Example: $X \sim \text{Exp}(1)$, $p = 0.9$

For the exponential distribution $F(x) = 1 - e^{-x}$ and density $f(x) = e^{-x}$:

$$q_{0.9} = F^{-1}(0.9) = -\ln(0.1) = \ln(10) \approx 2.303.$$

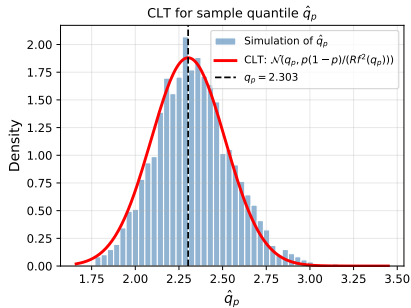
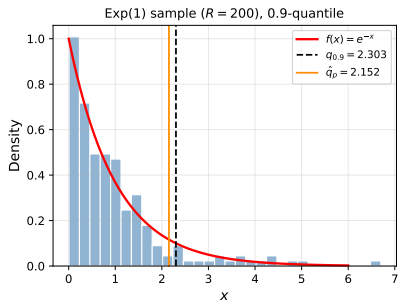
The density at the quantile is $f(q_{0.9}) = e^{-\ln 10} = 1/10 = 0.1$. Therefore:

$$\varsigma^2 = \frac{0.9 \times 0.1}{0.1^2} = \frac{0.09}{0.01} = 9.$$

For $R = 1,000$ replications: standard error $\varsigma/\sqrt{R} = 3/\sqrt{1000} \approx 0.0949$. The 95% CI half-width is $1.96 \times 0.0949 \approx 0.186$. Thus the CI is approximately 2.303 ± 0.186 .

To cross-check: this large variance ($\varsigma^2 = 9$) makes sense because the exponential density at its 90th percentile ($f(2.303) = 0.1$) is quite small, meaning the distribution is spread out there.

4.3 Asymptotic Normality of the Sample Quantile IV



4.3.1 Estimating $1/f(q_p)$ via Order Statistics I

The confidence interval for q_p requires $\varsigma^2 = p(1-p)/f(q_p)^2$, which means we need to estimate $1/f(q_p)$. The trick uses the quantile function $Q(p) = F^{-1}(p)$, whose derivative satisfies $Q'(p) = 1/f(Q(p))$. We approximate $Q'(p)$ by a finite-difference quotient using nearby order statistics.

Order-Statistic Estimator (Proposition 4.3.4, Eq. 3.3)

Choose a bandwidth $m = m_R$ (a positive integer depending on R) and define:

$$\hat{Q}'_{p,R} = \frac{R}{2m} \left(X_{(\lfloor pR \rfloor + m)} - X_{(\lfloor pR \rfloor - m + 1)} \right). \quad (3.3)$$

Intuition: The $(\lfloor pR \rfloor + m)$ -th order statistic is approximately $Q(p + m/R)$, and the $(\lfloor pR \rfloor - m + 1)$ -th is approximately $Q(p - m/R)$. Dividing the difference $Q(p + m/R) - Q(p - m/R)$ by $2m/R$ approximates the derivative $Q'(p) = 1/f(q_p)$.

Consistency (Proposition 4.3.4): If f has a continuous derivative near q_p and $m_R \rightarrow \infty$ with $m_R = o(R)$ as $R \rightarrow \infty$, then $\hat{Q}'_{p,R} \rightarrow Q'(p) = 1/f(q_p)$ with probability 1.

4.3.1 Estimating $1/f(q_p)$ via Order Statistics II

Choosing the bandwidth m_R :

- **Simple rule:** $m_R = \lfloor c\sqrt{R} \rfloor$ for some constant $c > 0$. This satisfies both $m_R \rightarrow \infty$ and $m_R = o(R)$, ensuring consistency. The default choice $m_R = \lfloor \sqrt{R} \rfloor$ (i.e. $c = 1$) is common in practice.
- **MSE-optimal (Bloch–Gastwirth):** Using $m_R = cR^{4/5}$ minimizes the asymptotic MSE of $\hat{Q}'_{p,R}$, achieving rate $R^{-2/5}$:

$$\mathbb{E}(\hat{Q}'_{p,R} - Q'(p))^2 \sim \frac{5}{4} \cdot 9^{-1/5} [Q'(p)]^{4/5} [Q''(p)]^{3/5} R^{-2/5}.$$

The rate $R^{-2/5}$ is slower than the $R^{-1/2}$ rate for the mean, reflecting the fundamental difficulty of density estimation at a point.

4.3.1 Estimating $1/f(q_p)$ via Order Statistics III

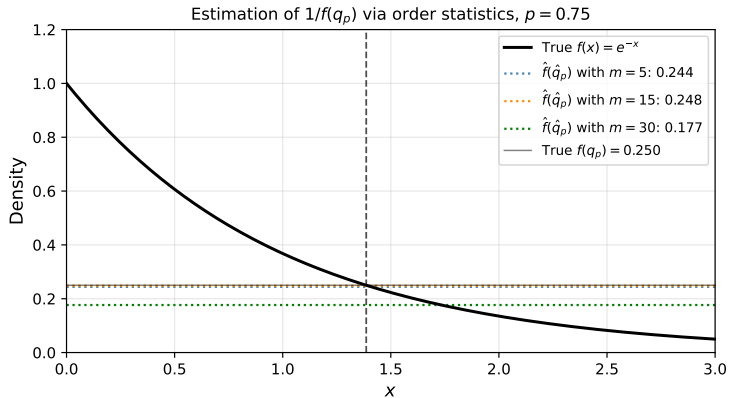
Confidence Interval for q_p (combining Prop. 4.3.1 and 4.3.4)

Substituting the consistent estimator $\hat{Q}'_{p,R}$ for $1/f(q_p)$:

$$\hat{q}_p \pm z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{R}} \hat{Q}'_{p,R}.$$

This CI has coverage converging to $1 - \alpha$ as $R \rightarrow \infty$.

4.3.1 Estimating $1/f(q_p)$ via Order Statistics IV



4.3 Python: Sample Quantile and Confidence Interval

```
import numpy as np
from scipy import stats

def quantile_ci(X_samples, p, m=None, alpha=0.05):
    """
    Estimate the p-quantile and compute a confidence interval.
    X_samples : 1D array of i.i.d. samples from F
    p          : quantile level in (0, 1)
    m          : bandwidth (default: floor(sqrt(R)))
    """
    R          = len(X_samples)
    X_sorted   = np.sort(X_samples)
    idx        = int(np.floor(p * R))      # index of sample quantile
    q_hat      = X_sorted[idx]            # sample quantile

    if m is None:
        m = max(1, int(np.floor(np.sqrt(R))))

    lo_idx = max(0, idx - m + 1)
    hi_idx = min(R - 1, idx + m)
    # Estimate  $Q'(p) = 1/f(q_p)$  via order-statistic finite differences
    Q_prime_hat = R * (X_sorted[hi_idx] - X_sorted[lo_idx]) / (2 * m)

    z_q      = stats.norm.ppf(1 - alpha / 2)
    margin   = z_q * np.sqrt(p * (1 - p) / R) * Q_prime_hat
    return q_hat, q_hat - margin, q_hat + margin
```

4.4.1 The Chernoff Bound I

Section 4.4 studies the comparison between absolute and relative errors more carefully, introducing a powerful exponential concentration inequality: the **Chernoff bound**. Unlike Chebyshev's inequality (polynomial decay $\sim 1/R$), the Chernoff bound gives *exponential decay* in R , making it enormously sharper for large samples.

4.4.1 The Chernoff Bound II

Proposition 4.4.1 – Chernoff Bound (Eqs. 4.1–4.3)

Let $S_R = Y_1 + \cdots + Y_R$ where $Y_i \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. For any $0 < \varepsilon < 1$:

$$\text{Upper tail: } \mathbb{P}(S_R \geq (1 + \varepsilon)Rp) \leq \exp\left(-\frac{\varepsilon^2 Rp}{2 + \varepsilon}\right) \leq \exp\left(-\frac{\varepsilon^2 Rp}{3}\right), \quad (4.1)$$

$$\text{Lower tail: } \mathbb{P}(S_R \leq (1 - \varepsilon)Rp) \leq \exp\left(-\frac{\varepsilon^2 Rp}{2}\right). \quad (4.2)$$

Combining both tails (Eq. 4.3):

$$\mathbb{P}(|S_R - Rp| \geq \varepsilon Rp) \leq 2 \exp\left(-\frac{\varepsilon^2 Rp}{3}\right).$$

4.4.1 The Chernoff Bound III

Proof of the upper tail (Markov exponential trick). For any $s > 0$, by Markov's inequality:
 $\mathbb{P}(S_R \geq x) \leq e^{-sx} \mathbb{E}[e^{sS_R}]$. For Bernoulli $_p$ variables: $\mathbb{E}[e^{sS_R}] = (pe^s + 1 - p)^R \leq e^{Rp(e^s - 1)}$ (using $1 + t \leq e^t$). Choosing $s = \ln(1 + \varepsilon)$ and applying $\ln(1 + \varepsilon) \geq \varepsilon^2/(2 + \varepsilon)$ gives the bound.

4.4.1 The Chernoff Bound IV

Corollary 4.4.2 – Required Replications for Relative Error

For $\hat{Y}_R = \frac{1}{R} \sum_{i=1}^R Y_i$ with $Y_i \sim \text{Bernoulli}(p)$, $\varepsilon, \delta \in (0, 1)$: if

$$R \geq \frac{3 \ln(2/\delta)}{\varepsilon^2 p},$$

then $\mathbb{P}(|\hat{Y}_R - p| \geq \varepsilon p) \leq \delta$ (Eq. 4.4). This controls *relative* error: notice the $1/p$ factor meaning rare events require far more replications.

4.4.1 The Chernoff Bound V

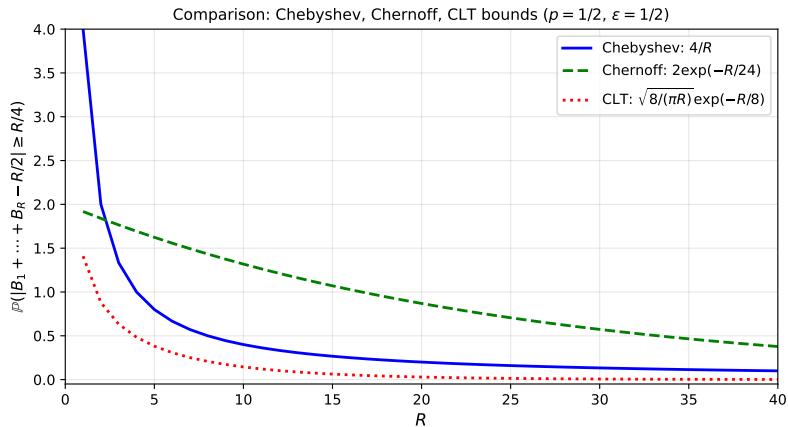
Example 4.4.3 – Three Bounds for Bernoulli(1/2)

We compare $\mathbb{P}(|B_1 + \dots + B_R - R/2| \geq R/4)$ under three approaches:

- **Chebyshev** (Eq. 4.5): $\leq 4/R$. Decreases like $1/R$.
- **Chernoff** (Eq. 4.6): $\leq 2e^{-R/24}$. Exponential decay.
- **CLT approximation** (Eq. 4.7): $\approx \sqrt{8/(\pi R)} e^{-R/8}$. The sharpest, also exponential.

For $R = 1,000$: Chebyshev gives $4/1000 = 0.004$; Chernoff gives $2e^{-41.7} \approx 1.7 \times 10^{-18}$; CLT gives $\approx 9 \times 10^{-55}$. The differences are astronomically large – Chebyshev is literally 10^{52} times larger than the CLT approximation.

4.4.1 The Chernoff Bound VI



4.4.2 Discussion on Absolute vs Relative Errors I

Consider i.i.d. Bernoulli(p) variables Y_i with $I = \mathbb{E}Y = p$, and fix precision $\varepsilon = \delta = 0.01$.

Absolute error first. From the CLT formula (1.7):

$$R^{\text{abs}} = z_{1-\delta/2}^2 \frac{\text{Var}(Y)}{\varepsilon^2} = 2.57^2 \cdot \frac{p(1-p)}{0.01^2}.$$

$p = I$	$\text{Var}(Y) = p(1-p)$	Required R^{abs}
0.1	0.09	$\approx 5,945$
0.01	0.0099	≈ 654
0.001	≈ 0.001	≈ 66

Counterintuitively, *smaller* p requires *fewer* replications for absolute error control! The reason:

$\text{Var}(Y) = p(1-p) \rightarrow 0$ as $p \rightarrow 0$. In the extreme: with $R = 1$, $\hat{Y}_1 = Y_1 \in \{0, 1\}$, and $\mathbb{P}(|\hat{Y}_1 - p| \leq 0.01) = \mathbb{P}(Y_1 = 0) = 1 - p \geq 0.99$ for $p \leq 0.01$ — this technically satisfies (E1) with $\varepsilon = \delta = 0.01$, but the estimate is either 0 or 1, nowhere near p !

The absolute error condition gives a misleading picture here: with high probability the error is indeed < 0.01 , but that is only because p itself is nearly 0.

4.4.2 Discussion on Absolute vs Relative Errors II

Relative error corrects this flaw. From Corollary 4.4.2, to control $\mathbb{P}(|\hat{Y}_R - p| \geq \varepsilon p) \leq \delta$:

$$R \geq \frac{3 \ln(2/\delta)}{\varepsilon^2 p}.$$

$p = I$	Required R (Chernoff, $\varepsilon = \delta = 0.01$)
0.1	$\approx 1.59 \times 10^6$
0.01	$\approx 1.59 \times 10^7$
0.001	$\approx 1.59 \times 10^8$
10^{-6}	$\approx 1.59 \times 10^{11}$

Now $R \propto 1/p$: the required sample size grows *inversely* with the rarity of the event. For $p = 10^{-6}$, one would need about 1.59×10^{11} replications — completely infeasible for naive Monte Carlo.

4.4.2 Discussion on Absolute vs Relative Errors III

Key Insight: Rare Events Require Variance Reduction

Controlling relative error for small I requires $R \propto 1/I$ replications. This is the fundamental hardness of rare-event simulation. Variance reduction techniques such as importance sampling (Chapter 5) replace Y with a lower-variance estimator, directly reducing the required R . Without them, naive Monte Carlo is hopeless for rare events.

4.4.3 Absolute vs Relative Error for Estimating π I

Since $\pi \approx 3.14$ is neither too small nor too large, both error criteria lead to manageable sample sizes. Let us quantify them carefully using the hit-or-miss estimator from Example 4.1.6:

$Y_j = 4\mathbf{1}(U_{2j-1}^2 + U_{2j}^2 \leq 1)$, $\mathbb{E}Y = \pi$, $\text{Var}(Y) = 16(\pi/4)(1 - \pi/4) \approx 2.6968$.

Fix $\varepsilon = \delta = 0.01$ (i.e. $z_{1-\delta/2} = z_{0.995} = 2.57$).

Absolute Error via CLT (Eq. 1.7)

$$R^{\text{abs}} = z_{1-\delta/2}^2 \frac{\text{Var}(Y)}{\varepsilon^2} = 2.57^2 \cdot \frac{2.6968}{0.01^2} \approx 1.78 \times 10^5.$$

Relative Error via Chernoff

Setting $\varepsilon = 0.01$ (1% relative accuracy) and using the indicator nature: $Y' = \mathbf{1}(U_1^2 + U_2^2 \leq 1)$ satisfies $\mathbb{E}Y' = \pi/4$, so:

$$R^{\text{rel},1} \geq \frac{3 \ln(2/0.01)}{0.01^2 \cdot (\pi/4)} \approx 5 \times 10^4.$$

4.4.3 Absolute vs Relative Error for Estimating π II

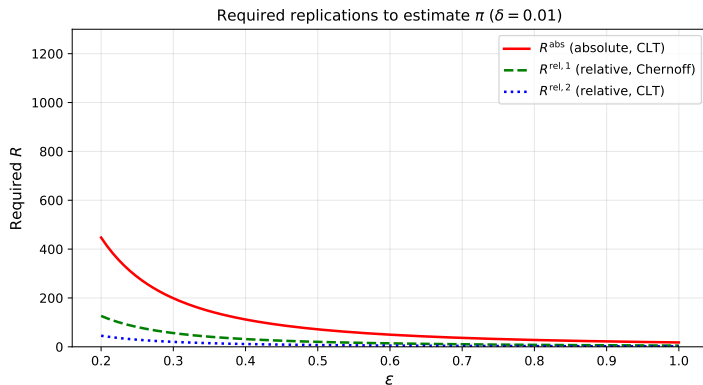
Relative Error via CLT

For $\hat{I} \approx \mathcal{N}(I, \text{Var}(Y)/R)$ and condition (E2'): $\mathbb{P}(|\hat{Y}_R - \pi| \leq 0.01\pi) \geq 0.99$:

$$R^{\text{rel},2} = z_{1-\delta/2}^2 \frac{\text{Var}(Y)}{\varepsilon^2 \pi^2} = 2.57^2 \cdot \frac{2.6968}{0.01^2 \cdot \pi^2} \approx 5 \times 10^5.$$

Since $\pi \approx 3.14$ is not small, the relative error requirement is only about $3\times$ harder than absolute error (CLT formula). The discrepancy between Chernoff and CLT for relative error arises because Chernoff applies to indicators and is a non-asymptotic bound, while CLT is asymptotic but accounts for the actual variance.

4.4.3 Absolute vs Relative Error for Estimating π III



4.4 Python: Comparing Error Bounds

```
import numpy as np
from scipy import stats

def R_absolute_clt(sigma2, epsilon, delta):
    """Required R for absolute error control via CLT."""
    z = stats.norm.ppf(1 - delta / 2)
    return z**2 * sigma2 / epsilon**2

def R_relative_chernoff(p, epsilon, delta):
    """Required R for relative error via Chernoff (Bernoulli only)."""
    return 3 * np.log(2 / delta) / (epsilon**2 * p)

def R_relative_clt(sigma2, I, epsilon, delta):
    """Required R for relative error via CLT."""
    z = stats.norm.ppf(1 - delta / 2)
    return z**2 * sigma2 / (epsilon**2 * I**2)

# Hit-or-miss estimator for pi
VarY = 16 * (np.pi / 4) * (1 - np.pi / 4) # approx 2.6968
I     = np.pi
eps   = 0.01
delt  = 0.01

R_abs  = R_absolute_clt(VarY, eps, delt)
R_rel1 = R_relative_chernoff(np.pi / 4, eps, delt) # Chernoff on indicator
R_rel2 = R_relative_clt(VarY, I, eps, delt)
```

4.5 Conclusions: The Fundamental Relationship Revisited I

The core result of this chapter is the **fundamental relationship** that governs the accuracy of all Monte Carlo estimators based on independent replications. Regardless of the specific estimation problem (mean, transformed mean, or quantile), the accuracy has the form:

The Fundamental Relationship (Eq. 5.1)

$$b = \frac{z_{1-\alpha/2} D}{\sqrt{R}},$$

where b is the CI half-width, R is the number of replications, $z_{1-\alpha/2}$ is the normal quantile at level $1 - \alpha$, and D is the variability constant depending on the problem:

- Estimating $I = \mathbb{E}Y$: $D = \sigma_Y = \sqrt{\text{Var}(Y)}$.
- Estimating $z = k(I)$ with $I = \mathbb{E}Y$ (1D): $D = |k'(I)| \sigma_Y$.
- Estimating $z = k(\mathbf{I})$ with $\mathbf{I} = \mathbb{E}\mathbf{Y}$ (multi-D): $D = \sqrt{\nabla k(\mathbf{I}) \boldsymbol{\Sigma}_Y (\nabla k(\mathbf{I}))^T}$.

In all cases the error decreases at the *Monte Carlo rate* $O(R^{-1/2})$, **independent of dimension** d . This is the key advantage of Monte Carlo over deterministic quadrature for high-dimensional problems.

4.5 Conclusions: The Fundamental Relationship Revisited II

Practical workflow for a simulation study:

- ① Run a *pilot simulation* of size m to estimate D (equivalently, σ_Y^2 or Σ_Y).
- ② Compute the required $R = z_{1-\alpha/2}^2 D^2 / b^2$ for the desired b and α .
- ③ Run R main replications; compute estimates and sample statistics.
- ④ Report the point estimate and confidence interval.

4.5 Conclusions: The Fundamental Relationship Revisited III

Summary of Confidence Intervals (for $\alpha = 0.05$, $z_{0.975} = 1.96$)

Quantity	Point Estimate	95% CI
$I = \mathbb{E}Y$	$\hat{Y}_R$	$\hat{Y}_R \pm \frac{1.96 \hat{S}_R}{\sqrt{R}}$
$k(I)$, $I = \mathbb{E}Y$	$k(\hat{Y}_R)$	$k(\hat{Y}_R) \pm \frac{1.96 \hat{S}_R k'(\hat{Y}_R) }{\sqrt{R}}$
$k(\mathbf{I})$, $\mathbf{I} = \mathbb{E}\mathbf{Y}$	$k(\hat{\mathbf{Y}}_R)$	$k(\hat{\mathbf{Y}}_R) \pm 1.96 \frac{\hat{\varsigma}}{\sqrt{R}}$
q_p (quantile)	$X_{(\lfloor pR \rfloor)}$	$\hat{q}_p \pm 1.96 \sqrt{\frac{p(1-p)}{R}} \hat{Q}'_{p,R}$

4.5 Conclusions: The Fundamental Relationship Revisited IV

Key Takeaways of Chapter 4

- The Monte Carlo rate $b = O(R^{-1/2})$ holds in any dimension – there is **no curse of dimensionality** for the statistical error.
- The CLT-based formula requires only $\approx 19\%$ as many replications as the (valid but conservative) Chebyshev bound.
- Transformed estimators $k(\hat{Y}_R)$ are generally biased with bias $O(R^{-1})$, which is dominated by standard deviation $O(R^{-1/2})$ for large R ; bias corrections are usually unnecessary.
- **Relative error** is far harder than absolute error when I is small; the required $R \propto 1/I$ making naive MC infeasible for rare events. Variance reduction (Chapter 5) is essential.
- For quantile estimation: $\varsigma^2 = p(1-p)/f(q_p)^2$ – the density value at the quantile must be estimated via order statistics, with bandwidth $m_R \sim \sqrt{R}$ (simple) or $m_R \sim R^{4/5}$ (MSE-optimal).
- The three shapes of confidence regions (ellipsoid, parallelogram, rectangle) trade off tightness for computational ease: the ellipsoid is smallest but requires eigendecomposition, the rectangle is largest but axis-aligned.

4.5 Python: Full Output Analysis Workflow

```
import numpy as np
from scipy import stats

def full_mc_analysis(simulate_Y, R_pilot=500, b=0.05, alpha=0.05, seed=42):
    """Complete Monte Carlo output analysis pipeline."""
    rng = np.random.default_rng(seed)

    # Step 1: Pilot simulation to estimate sigma_Y^2
    Y_pilot = np.array([simulate_Y(rng) for _ in range(R_pilot)])
    sigma2_hat = Y_pilot.var(ddof=1)
    print(f"Pilot sigma^2 estimate: {sigma2_hat:.4f}")

    # Step 2: Required R from the fundamental relationship
    z = stats.norm.ppf(1 - alpha / 2)
    R_main = int(np.ceil(z**2 * sigma2_hat / b**2))
    print(f"Required R (b={b}, alpha={alpha}): {R_main}")

    # Step 3: Main simulation
    Y_main = np.array([simulate_Y(rng) for _ in range(R_main)])
    Y_bar = Y_main.mean()
    S = Y_main.std(ddof=1)

    ci_lo = Y_bar - z * S / np.sqrt(R_main)
    ci_hi = Y_bar + z * S / np.sqrt(R_main)
    print(f"Estimate: {Y_bar:.4f}")
    print(f"95% CI : ({ci_lo:.4f}, {ci_hi:.4f})")
    return Y_bar, (ci_lo, ci_hi)
```

Chapter 4: Summary

Section 4.1 – Estimating $I = \mathbb{E}Y$:

- Sample mean $\hat{Y}_R = \frac{1}{R} \sum Y_j$: unbiased, consistent
- Chebyshev: $b_R = \sigma_Y / (\alpha^{1/2} \sqrt{R})$ (valid, loose)
- CLT: $b = z_{1-\alpha/2} \sigma_Y / \sqrt{R}$ (sharp, asymptotic)
- Required $R = z_{1-\alpha/2}^2 \sigma_Y^2 / b^2$
- CLT needs $\approx 19\%$ of Chebyshev's R
- Multivariate: ellipsoid, parallelogram, rectangle regions
- Sample covariance $\hat{\Sigma}_R^2$ for unknown Σ

Section 4.2 – Estimating $k(I)$:

- Delta method: $\varsigma^2 = k'(I)^2 \sigma_Y^2$
- Multi-D: $\varsigma^2 = \nabla k(\mathbf{I}) \Sigma (\nabla k)^T$
- Bias $O(R^{-1})$ negligible vs std dev $O(R^{-1/2})$
- $\text{MSE} = \text{Variance} + \text{Bias}^2$

Section 4.3 – Quantile estimation:

- $\hat{q}_p = X_{(\lfloor pR \rfloor)}$: consistent, CLT holds
- Asymptotic variance: $\varsigma^2 = p(1-p)/f(q_p)^2$
- Estimate $1/f(q_p)$ via order-stat differences (bandwidth m_R)
- Simple: $m_R \sim \sqrt{R}$; optimal: $m_R \sim R^{4/5}$

Section 4.4 – Error bounds:

- Chernoff: exponential bound for Bernoulli (vs polynomial Chebyshev)
- Absolute error: $R^{\text{abs}} \propto \text{Var}(Y)/\varepsilon^2$
- Relative error: $R^{\text{rel}} \propto \text{Var}(Y)/(\varepsilon^2 I^2)$
- Rare events ($I \ll 1$): $R \propto 1/I$ – need variance reduction

Monte Carlo rate: $b = O(R^{-1/2})$
independent of dimension d